



Statistical Inference for Generative Model Comparison

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2026 X-AGI
Beijing, China

From generative-model rankings to statistically supported comparisons.
Relative KL cancels an unknown entropy term; Edgeworth expansions refine inference with limited test data.

1 Why GenAI needs inference

Generative AI produces images and text by learning a **distribution**. A better-looking output or a lower point estimate does not, by itself, quantify a performance gap.



Generator 1



Generator 2

Visual motivation from the slides: similar-looking generated images can be hard to rank.

- **Statistical question:** does an observed difference reflect a population gap or finite-test-sample variability?
- **Challenge:** compare high-dimensional distributions when the target density p is unknown.

2 The idea: let the nuisance cancel

For fixed generative models $\hat{\mathbb{P}}_1, \hat{\mathbb{P}}_2$, define the absolute score

$$s(\hat{\mathbb{P}}_j) = -\text{KL}(\mathbb{P} \parallel \hat{\mathbb{P}}_j) = -\mathbb{E}_{\mathbb{P}} \log p(Y) + \mathbb{E}_{\mathbb{P}} \log \hat{p}_j(Y).$$

Compare the two scores directly.

$$\begin{aligned} \delta(\hat{\mathbb{P}}_1, \hat{\mathbb{P}}_2) &= -\text{KL}(\mathbb{P} \parallel \hat{\mathbb{P}}_1) + \text{KL}(\mathbb{P} \parallel \hat{\mathbb{P}}_2) \\ &= \mathbb{E}_{\mathbb{P}}[\log \hat{p}_1(Y) - \log \hat{p}_2(Y)]. \end{aligned}$$

The unknown entropy term cancels exactly. No estimate of the target density p is needed.

- $\delta > 0$: model 1 is closer to $\mathbb{P}$ in KL divergence. $\delta < 0$: model 2 is closer.
- **Why KL?** Up to scale, KL is the unique smooth f -divergence admitting this cancellation (paper, Proposition 1).
- **What is required?** i.i.d. test data, finite KL scores, and positive model likelihoods evaluable on the target support.

The same idea extends to conditional GenAI.

For question-answer pairs (X_i, Y_i) , average the conditional log-likelihood ratio:

$$\hat{\delta} = \frac{1}{n} \sum_{i=1}^n [\log \hat{p}_1(Y_i \mid X_i) - \log \hat{p}_2(Y_i \mid X_i)].$$

This compares how well the two models represent the answer distribution given a question.

Test data $\rightarrow$ paired log scores $\rightarrow$ relative KL + CI.

3 A simple estimator, valid inference

On $Y_i \stackrel{\text{i.i.d.}}{\sim} \mathbb{P}$, the relative score is estimated by a first-order U-statistic (a sample mean):

$$\hat{\delta} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} [\log \hat{p}_1(Y_i) - \log \hat{p}_2(Y_i)], \quad \mathbb{E} \hat{\delta} = \delta.$$

Let $V = \text{Var}[\log \hat{p}_1(Y) - \log \hat{p}_2(Y)]$. If $0 < V < \infty$,

$$\frac{\hat{\delta} - \delta}{\sqrt{\hat{V}/n_{\text{test}}}} \xrightarrow{d} \mathcal{N}(0, 1).$$

A $(1 - \alpha)$ -level asymptotic CI

$$\hat{\delta} \pm q_{1-\alpha/2} \sqrt{\hat{V}/n_{\text{test}}}.$$

An interval excluding zero identifies a performance difference with statistical confidence.

$\hat{V}$: empirical variance of the paired log scores; $q_u = \Phi^{-1}(u)$. Uncertainty is over the test sample, with the fitted models fixed.

4 EEs: refine the normal approximation

For limited test data, skewed log scores can make the CLT approximation inaccurate. **Edgeworth expansions (EEs)** retain higher-order information.

$$X_i = \log \hat{p}_1(Y_i) - \log \hat{p}_2(Y_i), \quad n = n_{\text{test}},$$

$$\mu = \delta, \quad \sigma^2 = V, \quad Z_n = \frac{\sqrt{n}(X_n - \mu)}{\sigma}.$$

Second-order distributional approximation

$$\begin{aligned} G(x) &= \Phi(x) - \phi(x) \left[\frac{\kappa_3}{6\sqrt{n}} H_2(x) \right. \\ &\quad \left. + \frac{1}{n} \left[\frac{\kappa_4}{24} H_3(x) + \frac{\kappa_3^2}{72} H_5(x) \right] \right]. \end{aligned}$$

$$\sup_{x \in \mathbb{R}} |F_n(x) - G(x)| = o(n^{-1}).$$

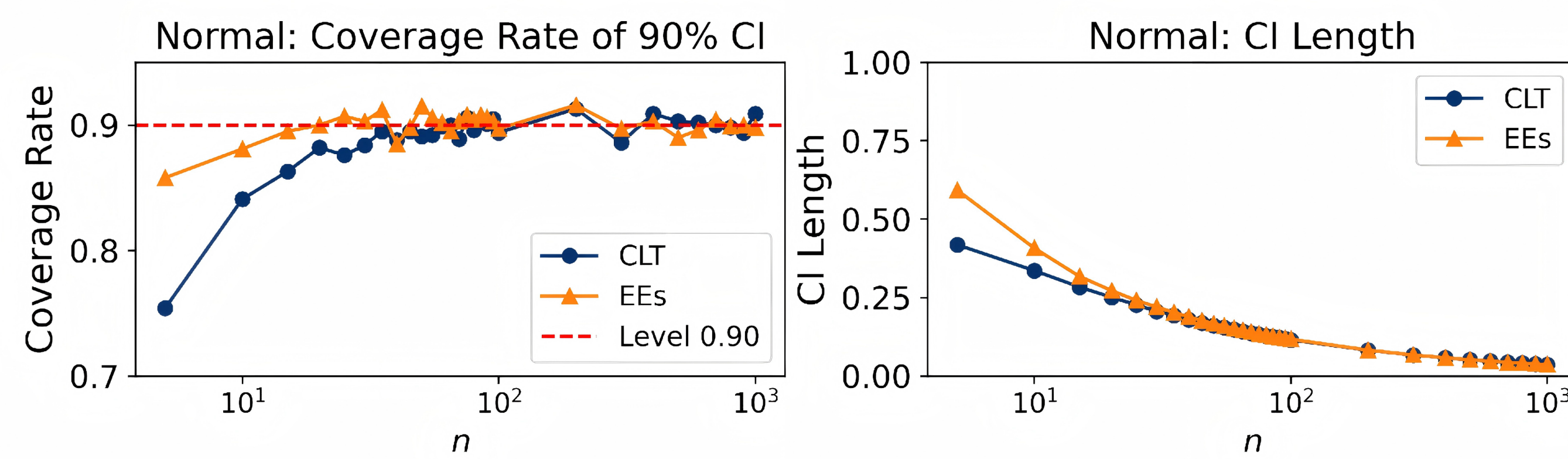
$\kappa_3 = m_3/\sigma^3$ (skewness), $\kappa_4 = m_4/\sigma^4 - 3$ (excess kurtosis), $m_r = \mathbb{E}[(X - \mu)^r]$. F_n : CDF of Z_n ; H_k : Hermite polynomial; $\phi = \Phi'$.

Conditions: $\mathbb{E}|X|^4 < \infty$, $0 < V < \infty$, and Cramér's condition $\limsup_{|t| \rightarrow \infty} |\psi_X(t)| < 1$.

Theory to practice: cumulants $\rightarrow$ characteristic-function expansion $\rightarrow$ Fourier inversion.

With unknown variance, use the studentized expansion G^* from the paper's appendix.

5 Small-sample calibration

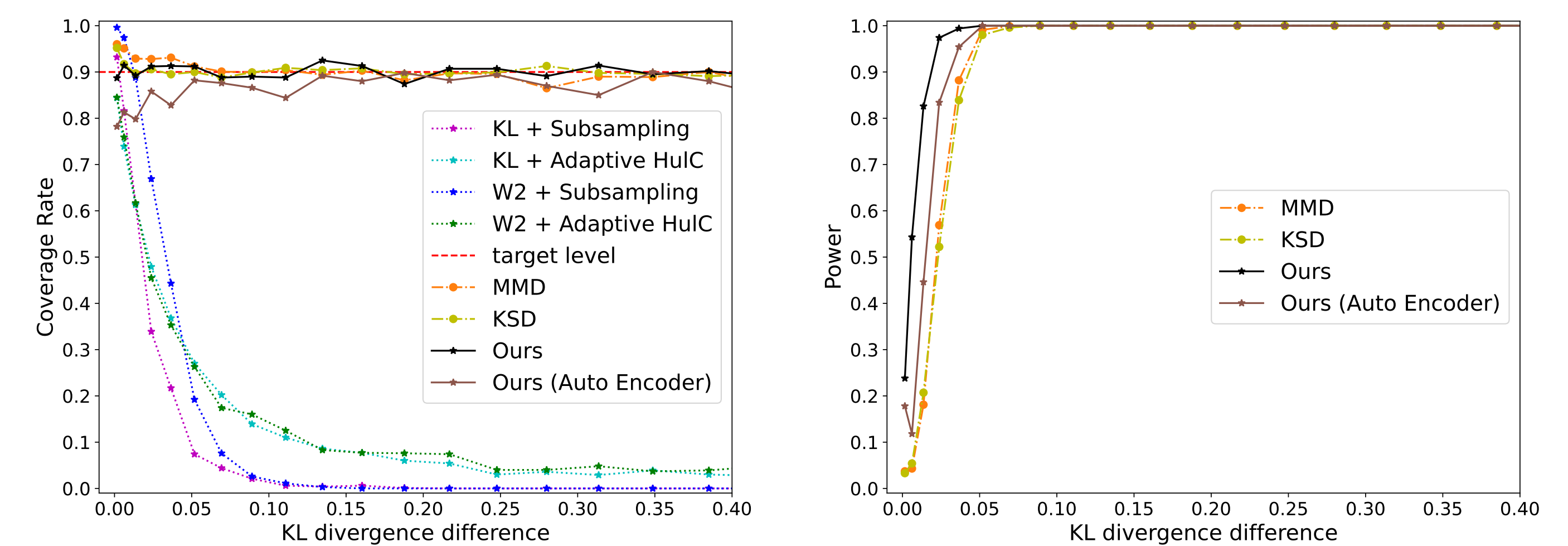


Gaussian simulation, $d = 10$, $\epsilon = 0.07$; nominal 90% CIs. Original camera-ready figure.

EEs improve coverage in this small-sample experiment, with wider intervals at the smallest n . The two methods approach each other as n increases.

6 Simulation: coverage and power

Gaussian generators with $d = 10$, $n = 1000$, 1000 replications; nominal 90% CIs.



Original simulation panels: coverage (left), power (right).

Direct relative-KL inference maintains near-nominal coverage. MMD and KSD also provide valid coverage, but have lower power in this experiment.

7 Real images: CIFAR-10

DDIM with S denoising steps; score relative to DDIM₁₀₀: $\delta(\hat{\mathbb{P}}_M, \hat{\mathbb{P}}_{\text{DDIM}_{100}})$.

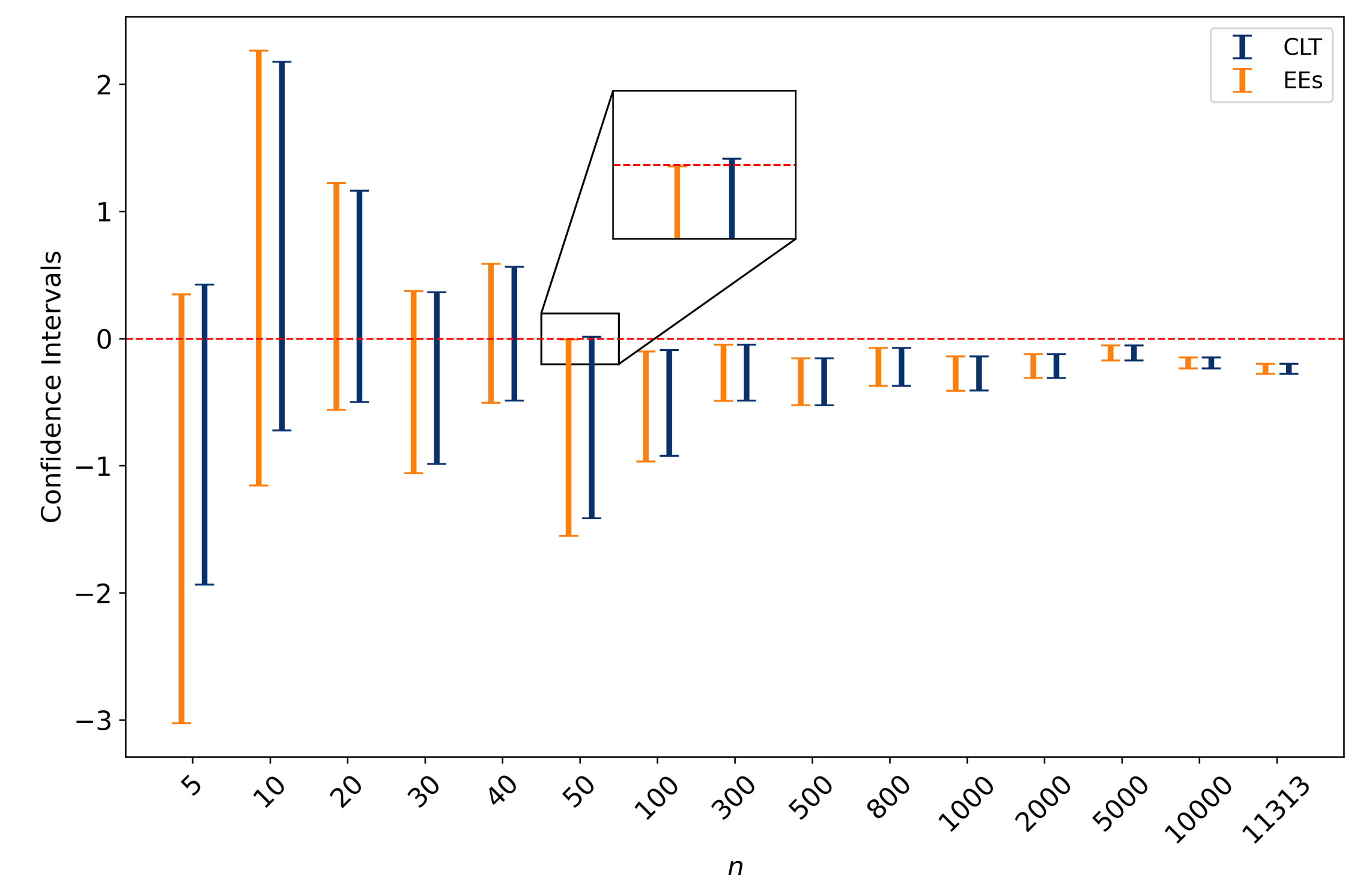
Model	FID ↓	CI of relative score
DDIM ₂₀	6.84	(−39.91, −38.70)
DDIM ₅₀	4.67	(−17.40, −16.63)
DDIM ₁₀₀	4.16	Reference

DDIM₁₀₀ is significantly better in this KL comparison, even when the FID gap against DDIM₅₀ is small.

8 Conditional GenAI: TriviaQA

Compare answer likelihoods given questions. Score relative to Llama3-8B: $\delta(\hat{\mathbb{P}}_M, \hat{\mathbb{P}}_{\text{Llama3-8B}})$.

Model	F1 ↑	CI
OPT-1.3B	7.47	(−7.19, −7.01)
Mistral-7B	12.24	(−0.22, −0.14)
Llama3-8B	36.34	Reference



Mistral-7B minus Llama3-8B. At $n = 50$, EEs excludes zero while CLT includes it in this reported subsample.

